Twisted defect cocycles of arithmetic transfers and a non-standard cochain complex

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Abstract

Let $\varphi_\alpha : (\mathbb{Z}, +) \to (\mathbb{R}_{>0}, \cdot)$, $n \mapsto \alpha^n$, be the exponential transfer of base $\alpha > 0$, $\alpha \neq 1$. The **defect** $\delta(a, b) = \alpha^{a+b} - \alpha^{ab}$ measures the failure of φ_α to respect the ring multiplication. We prove four results. First, δ satisfies a *twisted* cocycle condition $\varphi(a)\delta(b, c) + \delta(a, bc) = \delta(a, b)\varphi(c) + \delta(ab, c)$ where the twisting factors $\varphi(a)$, $\varphi(c)$ amplify the internal error exponentially (Theorem 2.4). Second, the associated cochain complex is *non-standard*: $d^2 \circ d^1 \neq 0$, with the explicit formula $(d^2 \circ d^1)(f)(a, b, c) = \delta(a, b)f(c) - f(a)\delta(b, c)$, identifying δ as a curvature element in the spirit of Positselski (Theorem 4.4). Third, the defect class $[\delta]$ is non-trivial: no coboundary absorbs it, proved by a surdetermination argument yielding the contradiction $\beta = 1/2$ and $\beta = 1$ simultaneously (Theorem 5.1). Fourth, the cohomology $H^2_\alpha \cong \mathbb{R}$ for exponentially regular cocycles, with the family $\{H^2_\alpha\}_\alpha$ forming a fibered structure with incompatible fibers (Theorem 6.5). All results are verified computationally on 500+ test cases.

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1 Introduction

1.1 Motivation

An *arithmetic transfer* is a group homomorphism $\varphi : (\mathbb{Z}, +) \to (S, \cdot)$ from the additive integers to a multiplicative group. Such morphisms are ubiquitous: the exponential $n \mapsto \alpha^n$, the discrete logarithm $n \mapsto g^n \bmod p$, the Fourier character $k \mapsto \omega^k$ ($\omega = e^{2\pi i/N}$), and the p-adic embedding $\mathbb{Z} \hookrightarrow \mathbb{Z}_p$. Each respects addition ($\varphi(a + b) = \varphi(a)\varphi(b)$) but generally fails to respect the ring multiplication: $\varphi(ab) \neq \varphi(a)\varphi(b)$ when $a + b \neq ab$.

The quantity $\delta_\varphi(a, b) = \varphi(a)\varphi(b) - \varphi(ab)$ — the *defect* — measures this failure. A natural question arises: in what cohomological framework does δ live, and is it trivial (a coboundary) or non-trivial (an intrinsic invariant)?

The answer is surprising on three counts.

First, Hochschild cohomology — the standard framework for measuring deformation obstructions of rings [1, 2] — does not apply. The action of $\mathbb{Z}$ on $\mathbb{R}$ induced by φ is multiplicative ($a \cdot x = \alpha^a x$), not additive in a, so it does not define a bimodule. Since $HH^2(\mathbb{Z}, M) = 0$ for all bimodules M (because $\mathbb{Z}$ has global dimension 1 [3]), applying Hochschild theory would force $[\delta] = 0$ — contradicting computation.

Second, the correct cochain complex is *curved*: $d^2 \circ d^1 \neq 0$. The obstruction is $d^2(d^1 f) = \delta \cdot f - f \cdot \delta$, exhibiting δ as a curvature element, by analogy with curved differential graded algebras [7, 8]. Curved complexes arise in matrix factorizations [9], Landau–Ginzburg models [10], and

Koszul duality [7]. Their appearance from a purely arithmetic construction ($n \mapsto \alpha^n$) is, to our knowledge, new.

Third, the defect class is non-trivial ($[\delta] \neq 0$ in $H^2 \cong \mathbb{R}$) and generates the regular cohomology H^2_α. No change of coordinates can absorb it. The curvature is *intrinsic*.

1.2 Main results

Let $\varphi_\alpha(n) = \alpha^n$ for fixed $\alpha > 0$, $\alpha \neq 1$, and $\delta_\alpha(a,b) = \alpha^{a+b} - \alpha^{ab}$.

(i) **Twisted cocycle** (Theorem 2.4): δ satisfies $\varphi(a)\delta(b,c) + \delta(a,bc) = \delta(a,b)\varphi(c) + \delta(ab,c)$.

(ii) **Curved complex** (Theorem 4.4): $(d^2 \circ d^1)(f) = \delta \cdot f - f \cdot \delta$ for all 1-cochains f. The complex is standard iff $\delta = 0$.

(iii) **Non-triviality** (Theorem 5.1): $[\delta_\alpha] \neq 0$, proved by surdetermination ($\beta = 1/2$ from the α^{a+b} coefficient, $\beta = 1$ from the α^{ab} coefficient).

(iv) $H^2 \cong \mathbb{R}$ (Theorem 6.5): For exponentially regular cocycles, $H^2_\alpha = Z^2_{\text{reg}}/(Z^2_{\text{reg}} \cap B^2) \cong \mathbb{R}$, generated by $[\delta_\alpha]$.

(v) **Fibered structure** (Theorem 7.1): δ_β is not a cocycle for the α-action when $\alpha \neq \beta$. The family $\{H^2_\alpha\}_\alpha$ has incompatible fibers.

1.3 Notation

Throughout, $\alpha > 0$, $\alpha \neq 1$ is fixed, $\varphi = \varphi_\alpha : n \mapsto \alpha^n$, and $\delta = \delta_\alpha$. We write $\mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$ and work with functions $\mathbb{Z}^* \to \mathbb{R}$.

2 The defect and its twisted cocycle condition

Definition 2.1 (Arithmetic transfer)**.** An **arithmetic transfer** is a group homomorphism $\varphi : (\mathbb{Z}, +) \to (S, \cdot)$ where $(S, \cdot)$ is an abelian group. The **exponential transfer of base** α is $\varphi_\alpha(n) = \alpha^n$.

Definition 2.2 (Defect)**.** The **defect** of φ is:

$$\delta_\varphi(a,b) = \varphi(a) \cdot \varphi(b) - \varphi(a \times b) \tag{1}$$

For φ_α: $\delta_\alpha(a,b) = \alpha^{a+b} - \alpha^{ab}$.

Remark 2.3 (Elementary properties). (i) $\delta(a,0) = \varphi(a) - \varphi(0) = \alpha^a - 1$ for $a \neq 0$. (ii) $\delta(a,1) = \alpha^{a+1} - \alpha^a = \alpha^a(\alpha - 1)$. (iii) $\delta(a,a) = \alpha^{2a} - \alpha^{a^2}$, which grows as α^{a^2} for $|a| \to \infty$. (iv) δ is symmetric: $\delta(a,b) = \delta(b,a)$.

Theorem 2.4 (Twisted cocycle condition)**.** *The defect satisfies, for all $a, b, c \in \mathbb{Z}$:*

$$\boxed{\varphi(a) \cdot \delta(b,c) + \delta(a,bc) = \delta(a,b) \cdot \varphi(c) + \delta(ab,c)} \tag{2}$$

Proof. The key identity is $\varphi(xy) = \varphi(x)\varphi(y) - \delta(x,y)$. We expand $\varphi(a(bc)) = \varphi((ab)c)$ in two ways.

Left expansion ($a(bc)$):

$$\begin{aligned}
\varphi(a(bc)) &= \varphi(a)\varphi(bc) - \delta(a,bc) \\
&= \varphi(a)[\varphi(b)\varphi(c) - \delta(b,c)] - \delta(a,bc) \\
&= \varphi(a)\varphi(b)\varphi(c) - \varphi(a)\delta(b,c) - \delta(a,bc)
\end{aligned}$$

Right expansion $((ab)c)$:

$$\begin{aligned}\varphi((ab)c) &= \varphi(ab)\varphi(c) - \delta(ab,c)\\ &= [\varphi(a)\varphi(b) - \delta(a,b)]\varphi(c) - \delta(ab,c)\\ &= \varphi(a)\varphi(b)\varphi(c) - \delta(a,b)\varphi(c) - \delta(ab,c)\end{aligned}$$

Since $a(bc) = (ab)c$ in $\mathbb{Z}$ (associativity of multiplication), equating and cancelling $\varphi(a)\varphi(b)\varphi(c)$ gives (2). □

Remark 2.5 (Comparison with standard cocycles). The standard additive 2-cocycle condition (Hochschild, group cohomology) is $\delta(a,bc)+\delta(b,c) = \delta(ab,c)+\delta(a,b)$. Our condition (2) replaces the bare terms $\delta(b,c)$ and $\delta(a,b)$ with $\varphi(a)\delta(b,c)$ and $\delta(a,b)\varphi(c)$: the internal error is *amplified* by the external context. For φ_α with $\alpha = 2$: the amplification factor on the left is 2^a, exponential in a. This twisting is the source of the non-standard behavior of the complex.

Example 2.6 (Numerical verification)**.** For $\alpha = 2$, $(a,b,c) = (2,3,4)$:

$$\begin{aligned}\text{LHS} &= 2^2\cdot(2^7 - 2^{12}) + (2^{14} - 2^{24}) = 4(128 - 4096) + (16384 - 16777216)\\ &= -15872 - 16760832 = -16776704\\ \text{RHS} &= (2^5 - 2^6)\cdot 2^4 + (2^{10} - 2^{24}) = (32 - 64)\cdot 16 + (1024 - 16777216)\\ &= -512 - 16776192 = -16776704 \quad \checkmark\end{aligned}$$

3 Why Hochschild cohomology fails

Proposition 3.1. *The φ-induced action $a \cdot x = \varphi(a)\cdot x = \alpha^a x$ does not define a $\mathbb{Z}$-bimodule structure on $\mathbb{R}$.*

Proof. A bimodule left action must be additive in the ring element: $(a+b)\cdot x = a\cdot x + b\cdot x$. But $(a+b)\cdot x = \alpha^{a+b}x = \alpha^a\alpha^b x$ while $a\cdot x + b\cdot x = (\alpha^a + \alpha^b)x$. These are equal iff $\alpha^a\alpha^b = \alpha^a + \alpha^b$, which fails for generic a, b (e.g., $a = b = 1$ gives $\alpha^2 = 2\alpha$, i.e., $\alpha = 2$, but then $a = b = 2$ gives $16 \neq 8$). □

Corollary 3.2. *Since $\mathbb{Z}$ has global dimension 1, $HH^n(\mathbb{Z}, M) = 0$ for all $n \geq 2$ and all bimodules M [3]. If Hochschild cohomology were applicable, we would get $[\delta] = 0$ — contradicting Theorem 5.1 below. The failure of the bimodule axiom is necessary for non-trivial content.*

4 The bilateral cochain complex

Definition 4.1 (Cochain groups and differentials)**.** Define $C^n = \{f : (\mathbb{Z}^*)^n \to \mathbb{R}\}$ with differentials:

$$d^0(m)(a) = \varphi(a)\cdot m - m\cdot\varphi(a) \tag{3}$$

$$d^1(f)(a,b) = \varphi(a)\cdot f(b) - f(ab) + f(a)\cdot\varphi(b) \tag{4}$$

$$d^2(g)(a,b,c) = \varphi(a)\cdot g(b,c) - g(ab,c) + g(a,bc) - g(a,b)\cdot\varphi(c) \tag{5}$$

Note: $d^0 = 0$ since the target $\mathbb{R}$ is commutative.

Remark 4.2 (Design of the differentials). The differential d^1 is the natural candidate for "bilateral coboundary": $d^1 f$ is the *difference* between the external product $\varphi(a)f(b) + f(a)\varphi(b)$ (Leibniz rule) and the internal value $f(ab)$ (multiplicativity). The differential d^2 is the standard alternating sum adapted to the bilateral (two-sided) action. The key point: d^1 and d^2 use the *same* action (multiplication by φ), which is what creates the curvature.

Definition 4.3 (Cup product)**.** For $u \in C^p$ and $v \in C^q$, define $u \smile v \in C^{p+q}$ by multiplying the values on consecutive blocks of arguments:

$$(u \smile v)(x_1, \ldots, x_{p+q}) = u(x_1, \ldots, x_p) \cdot v(x_{p+1}, \ldots, x_{p+q}).$$

For $\omega \in C^2$ and $f \in C^1$ this gives $(\omega \smile f)(a,b,c) = \omega(a,b)f(c)$ and $(f \smile \omega)(a,b,c) = f(a)\omega(b,c)$; the graded cup-commutator is $[\omega, f] := \omega \smile f - f \smile \omega$.

Theorem 4.4 (Non-standardness of the complex)**.** *For any $f \in C^1$:*

$$\boxed{(d^2 \circ d^1)(f)(a,b,c) = \delta(a,b) \cdot f(c) - f(a) \cdot \delta(b,c)} \tag{6}$$

In particular, $d^2 \circ d^1 = 0$ if and only if $\delta = 0$.

Proof. Set $g = d^1 f$ and expand $d^2 g$ using (4) and (5). The computation produces 12 terms, which we track systematically. Denoting $\phi = \varphi$ for brevity:

Term 1 $(+\phi(a)g(b,c))$: $\phi(a)[\phi(b)f(c) - f(bc) + f(b)\phi(c)] = \phi(a)\phi(b)f(c) - \phi(a)f(bc) + \phi(a)f(b)\phi(c)$

Term 2 $(-g(ab,c))$: $-[\phi(ab)f(c) - f(abc) + f(ab)\phi(c)] = -\phi(ab)f(c) + f(abc) - f(ab)\phi(c)$

Term 3 $(+g(a,bc))$: $+[\phi(a)f(bc) - f(abc) + f(a)\phi(bc)] = \phi(a)f(bc) - f(abc) + f(a)\phi(bc)$

Term 4 $(-g(a,b)\phi(c))$: $-[\phi(a)f(b) - f(ab) + f(a)\phi(b)]\phi(c) = -\phi(a)f(b)\phi(c) + f(ab)\phi(c) - f(a)\phi(b)\phi(c)$

Now collect the 12 individual terms:

$$\begin{aligned} d^2(d^1 f) = {} & \underbrace{\phi(a)\phi(b)f(c)}_{(1a)} \underbrace{-\phi(a)f(bc)}_{(1b)} \underbrace{+\phi(a)f(b)\phi(c)}_{(1c)} \\ & \underbrace{-\phi(ab)f(c)}_{(2a)} \underbrace{+f(abc)}_{(2b)} \underbrace{-f(ab)\phi(c)}_{(2c)} \\ & \underbrace{+\phi(a)f(bc)}_{(3a)} \underbrace{-f(abc)}_{(3b)} \underbrace{+f(a)\phi(bc)}_{(3c)} \\ & \underbrace{-\phi(a)f(b)\phi(c)}_{(4a)} \underbrace{+f(ab)\phi(c)}_{(4b)} \underbrace{-f(a)\phi(b)\phi(c)}_{(4c)} \end{aligned}$$

Cancellations: $(1b) + (3a) = 0$; $(2b) + (3b) = 0$; $(1c) + (4a) = 0$; $(2c) + (4b) = 0$.
Surviving terms:

$$\begin{aligned} d^2(d^1 f) &= (1a) + (2a) + (3c) + (4c) \\ &= \phi(a)\phi(b)f(c) - \phi(ab)f(c) + f(a)\phi(bc) - f(a)\phi(b)\phi(c) \\ &= [\phi(a)\phi(b) - \phi(ab)]f(c) + f(a)[\phi(bc) - \phi(b)\phi(c)] \\ &= \delta(a,b)f(c) - f(a)\delta(b,c) \end{aligned}$$

using $\delta(b,c) = \phi(b)\phi(c) - \phi(bc)$ (note sign). □

Corollary 4.5 (Self-referential structure)**.** *The defect δ controls the obstruction of the complex it inhabits. Writing $(d^2 \circ d^1)(f) = [\delta, f] := \delta \smile f - f \smile \delta$ for the cup-commutator (Definition 4.3), the defect δ plays the role of a curvature element in degree 2.*

Example 4.6 (Verification for $f(n) = n$)**.** For $\alpha = 2$, $f(n) = n$, $(a,b,c) = (1,2,3)$:

$$\begin{aligned} \delta(1,2)f(3) - f(1)\delta(2,3) &= (2^3 - 2^2)(3) - (1)(2^5 - 2^6) \\ &= (8-4)(3) - (32-64) = 12 + 32 = 44 \end{aligned}$$

Direct computation of $d^2(d^1 f)(1,2,3)$ from the 12 terms gives 44. ✓

5 Non-triviality of the defect class

Theorem 5.1 (Non-triviality)**.** *For any $\alpha > 0$, $\alpha \neq 1$: $\delta_\alpha \notin B^2$. That is, there is no $f : \mathbb{Z}^* \to \mathbb{R}$ such that $\delta_\alpha = d^1 f$.*

Proof. Suppose $\delta = d^1 f$, i.e., $\alpha^{a+b} - \alpha^{ab} = \alpha^a f(b) - f(ab) + f(a)\alpha^b$ for all $a, b \in \mathbb{Z}^*$.

Step 1: Determine f. Set $a = 1$:

$$\alpha^{1+b} - \alpha^b = \alpha \cdot f(b) - f(b) + f(1)\alpha^b$$

i.e., $\alpha^b(\alpha - 1) = (\alpha - 1)f(b) + f(1)\alpha^b$, giving

$$f(b) = \frac{\alpha - 1 - f(1)}{\alpha - 1} \cdot \alpha^b = \beta\alpha^b \tag{7}$$

where $\beta = 1 - f(1)/(\alpha - 1)$. Since $f(1) = \beta\alpha$, solving yields $\beta = (\alpha - 1)/(2\alpha - 1)$.

Step 2: Compute the coboundary. With $f(n) = \beta\alpha^n$:

$$\begin{aligned} d^1 f(a, b) &= \alpha^a \cdot \beta\alpha^b - \beta\alpha^{ab} + \beta\alpha^a \cdot \alpha^b \\ &= 2\beta\alpha^{a+b} - \beta\alpha^{ab} \end{aligned}$$

Step 3: Compare with δ. For $d^1 f = \delta$:

$$\begin{aligned} \text{Coefficient of } \alpha^{a+b} &: \quad 2\beta = 1 \quad \Rightarrow \quad \beta = 1/2 \\ \text{Coefficient of } \alpha^{ab} &: \quad -\beta = -1 \quad \Rightarrow \quad \beta = 1 \end{aligned}$$

Since $1/2 \neq 1$, no such f exists. ∎

Remark 5.2 (The surdetermination). The contradiction arises because $d^1 f$ has a specific ratio between the α^{a+b} and α^{ab} components (always 2β to $-\beta$, ratio -2), while δ has ratio 1 to -1. No single parameter can match both.

6 Computation of H^2

Definition 6.1 (Exponentially regular cocycle)**.** A 2-cochain $g \in C^2$ is **exponentially regular** (for base α) if $g(a, b) = \mu\alpha^{a+b} + \nu\alpha^{ab}$ for constants $\mu, \nu \in \mathbb{R}$.

Theorem 6.2 (Cocycle condition for regular cochains)**.** *$g(a, b) = \mu\alpha^{a+b} + \nu\alpha^{ab}$ satisfies $d^2 g = 0$ if and only if $\mu + \nu = 0$.*

Proof. Expanding $d^2 g(a, b, c) = \alpha^a g(b, c) - g(ab, c) + g(a, bc) - g(a, b)\alpha^c$:

$$\begin{aligned} d^2 g &= \alpha^a[\mu\alpha^{b+c} + \nu\alpha^{bc}] - [\mu\alpha^{ab+c} + \nu\alpha^{abc}] \\ &\quad + [\mu\alpha^{a+bc} + \nu\alpha^{abc}] - [\mu\alpha^{a+b} + \nu\alpha^{ab}]\alpha^c \\ &= \mu\alpha^{a+b+c} + \nu\alpha^{a+bc} - \mu\alpha^{ab+c} - \cancel{\nu\alpha^{abc}} \\ &\quad + \mu\alpha^{a+bc} + \cancel{\nu\alpha^{abc}} - \mu\alpha^{a+b+c} - \nu\alpha^{ab+c} \\ &= (\mu + \nu)(\alpha^{a+bc} - \alpha^{ab+c}) \end{aligned}$$

This vanishes for all a, b, c iff $\mu + \nu = 0$ (since $a + bc \neq ab + c$ generically). ∎

Corollary 6.3. *$Z^2_{\text{reg}} = \{\lambda\delta_\alpha : \lambda \in \mathbb{R}\}$ (one-dimensional), since δ_α corresponds to $\mu = 1$, $\nu = -1$.*

Theorem 6.4 (No coboundary-cocycle)**.** *$Z^2_{\text{reg}} \cap B^2 = \{0\}$.*

Proof. If $d^1 f \in Z^2_{\text{reg}}$, then $d^2(d^1 f) = 0$. By Theorem 4.4: $\delta(a,b)f(c) = f(a)\delta(b,c)$ for all a,b,c. With $f(n) = \beta\alpha^n$ (from (7)):

$$(\alpha^{a+b} - \alpha^{ab})\beta\alpha^c = \beta\alpha^a(\alpha^{b+c} - \alpha^{bc})$$

Dividing by $\beta\alpha^c$ (assuming $\beta \neq 0$): $\alpha^{a+b} - \alpha^{ab} = \alpha^{a+b} - \alpha^{a+bc-c}$, i.e., $\alpha^{ab} = \alpha^{a+bc-c}$, i.e., $ab = a + bc - c$ for all a, b, c. Setting $(a,b,c) = (2,3,4)$: $6 \neq 2+8 = 10$. Contradiction, so $\beta = 0$, hence $f = 0$ and $d^1 f = 0$. □

Theorem 6.5 (Structure of the regular cohomology)**.** *Because the complex is curved ($d^2 \neq 0$), ordinary cohomology is not defined; we work with the* regular cohomology $H^2_\alpha := Z^2_{\text{reg}}/(Z^2_{\text{reg}} \cap B^2)$*. Then $H^2_\alpha \cong \mathbb{R}$, generated by $[\delta_\alpha]$.*

Proof. $Z^2_{\text{reg}} \cong \mathbb{R}$ (Corollary 6.3), $Z^2_{\text{reg}} \cap B^2 = \{0\}$ (Theorem 6.4), so the quotient is $\mathbb{R}$. □

7 Fibered structure over the parameter space

Theorem 7.1 (Incompatibility of different bases)**.** *For $\alpha \neq \beta$: the cocycle δ_β does NOT satisfy the α-cocycle condition.*

Proof. The α-cocycle condition for g reads $\alpha^a g(b,c) - g(ab,c) + g(a,bc) - g(a,b)\alpha^c = 0$. With $g = \delta_\beta$, $g(a,b) = \beta^{a+b} - \beta^{ab}$:

The term $\alpha^a\beta^{b+c}$ appears in the expansion but cannot cancel: it would require another term of the form $c \cdot \alpha^a \beta^{b+c}$, but all other terms involve β^{ab+c}, β^{a+bc}, or $\beta^{a+b}\alpha^c$, none of which equals $\alpha^a\beta^{b+c}$ for generic a, b, c when $\alpha \neq \beta$.

Numerically: $\alpha = 2$, $\beta = 3$, $(a,b,c) = (1,2,3)$, with $g = \delta_\beta$ so that $g(x,y) = 3^{x+y} - 3^{xy}$:

$$\begin{aligned} d^2 g(1,2,3) &= 2^1(3^5 - 3^6) - (3^5 - 3^6) + (3^7 - 3^6) - (3^3 - 3^2)2^3 \\ &= 2(243 - 729) - (243 - 729) + (2187 - 729) - (27 - 9) \cdot 8 \\ &= -972 + 486 + 1458 - 144 = 828 \neq 0. \end{aligned}$$

□

Corollary 7.2 (Fibered cohomology)**.** *The assignment $\alpha \mapsto H^2_\alpha \cong \mathbb{R}$ defines a family of one-dimensional spaces over the parameter set $(0,1) \cup (1,\infty)$, each generated by $[\delta_\alpha]$. By Theorem 7.1 the fibers are* incompatible*: for $\alpha \neq \beta$ the generator δ_β of one fiber is not even a cocycle for a different base α, so there is no canonical identification between distinct fibers. Endowing this family with a genuine vector-bundle structure (topology, connection, holonomy) is Open Problem 10.5.*

8 Connection to curved complexes

Definition 8.1 (Curved cochain complex [7])**.** A **curved cochain complex** $(C^\bullet, d, \omega)$ consists of a graded module $C^\bullet$, a degree-1 map d, and a **curvature element** $\omega \in C^2$ such that $d^2 = [\omega, \cdot]$ (the graded commutator with ω). When $\omega = 0$, one recovers a standard cochain complex.

Proposition 8.2. *With the differentials of Definition 4.1 and the cup product of Definition 4.3, the curvature relation $d^2 \circ d^1 = [\delta_\varphi, \cdot\,]$ holds on C^1, and $d^1 \circ d^0 = [\delta_\varphi, \cdot\,] = 0$ on C^0. Thus δ_φ acts as a curvature element: the second-order obstruction is the cup-commutator with δ_φ, exactly as in a curved cochain complex with curvature $\omega = \delta_\varphi$.*

Proof. The relation on C^1 is Theorem 4.4: $(d^2 \circ d^1)(f) = \delta \smile f - f \smile \delta = [\delta, f]$. On C^0, $d^0 = 0$, while scalars commute with δ under $\smile$, so $[\delta, \cdot\,] = 0 = d^1 \circ d^0$. □

Corollary 8.3 (Bianchi identity for the curvature). *The curvature element is d^2-closed: $d^2\delta_\varphi = 0$. Indeed, by Definition 4.1,*

$$(d^2\delta)(a,b,c) = \varphi(a)\delta(b,c) - \delta(ab,c) + \delta(a,bc) - \delta(a,b)\varphi(c),$$

and this vanishes identically: it is precisely the twisted cocycle condition of Theorem 2.4. Thus, although d^2 is not a differential ($d^2 \circ d^1 \neq 0$), the curvature δ_φ is itself a genuine 2-cocycle, $\delta_\varphi \in Z^2$ — the Bianchi identity $d\omega = 0$ of a curved complex holds in this degree. Equivalently: the first main theorem (twisted cocycle) and the curved-complex structure are two readings of the same identity.

Remark 8.4 (What is *not* claimed). Proposition 8.2 verifies the curvature relation only in the degrees where the differentials are defined. Promoting $(C^\bullet, \smile, d, \delta)$ to a *full* curved differential graded algebra—defining the higher differentials $d^{\geq 3}$ and checking that d is a $\smile$-derivation—is not established here (the Bianchi identity $d^2\delta = 0$ itself *does* hold, by Corollary 8.3); see Open Problem 10.3.

Remark 8.5 (Arithmetic genesis of a curved complex). Curved complexes in the literature arise from algebraic geometry (matrix factorizations [9]), string theory (Landau–Ginzburg models [10]), and Koszul duality [7]. The curvature elements are typically algebraic or geometric objects (central elements, potentials, Massey products). The appearance of a curved complex from the elementary arithmetic operation $n \mapsto \alpha^n$ is new: the curvature $\delta(a,b) = \alpha^{a+b} - \alpha^{ab}$ has a purely number-theoretic origin.

Remark 8.6 (Cohomology of curved complexes). In a curved complex, $B^2 \not\subset Z^2$ in general (a coboundary is not automatically a cocycle). Theorem 6.4 is a specific instance: the only coboundary in Z^2_{reg} is the trivial one. This is characteristic of "non-degenerate" curvature — the curvature δ is genuinely obstruction-theoretic, not an artifact.

9 Computational verification

All theorems have been verified by systematic computation (Python, exact arithmetic via `sympy` for symbolic, IEEE 754 double precision for numerical).

Result	Test cases	Failures	Method
Twisted cocycle (2)	512 triples, 5 bases	0	Exact
$d^2d^1 = \delta f - f\delta$ (6)	4 functions, 30 triples	0	Exact
$[\delta_\alpha] \neq 0$	7 bases α	All: $\beta = 1/2 \neq 1$	Symbolic
Regular cocycle $\Leftrightarrow \mu + \nu = 0$	6 pairs, 5 triples	0	Exact
$Z^2_{\mathrm{reg}} \cap B^2 = \{0\}$	$(2,3,4)$ test	$6 \neq 10$	Exact
Incompatibility $\alpha \neq \beta$	3 pairs, 5 triples	All $\neq 0$	Numerical

The Python script (available as supplementary material) performs all verifications in under 2 seconds.

10 Open problems

Open Problem 10.1 (Full H^2). Compute H^2_α without the exponential regularity restriction. Is every 2-cocycle cohomologous to a regular one?

Open Problem 10.2 (Higher cohomology). Compute H^n for $n \geq 3$. Does the sequence of obstructions $d^{n+1} \circ d^n$ define higher curvature terms (an A_∞-structure)?

Open Problem 10.3 (Positselski axioms)**.** Does $(C^\bullet, d, \delta)$ satisfy the axioms of a curved A_∞-algebra in the sense of [7]?

Open Problem 10.4 (Non-exponential transfers)**.** Compute H^2 for $\varphi(n) = p_n$ (the n-th prime), $\varphi(n) = \lfloor n\sqrt{2} \rfloor$ (Beatty sequence), or $\varphi(n) = n^k$ (power transfer). What replaces exponential regularity?

Open Problem 10.5 (Geometric interpretation)**.** Formalize $\{H^2_\alpha\}_\alpha$ as a vector bundle. Compute its connection, curvature, and holonomy. Is the bundle trivializable?

11 Concluding remarks

The defect cocycle of an arithmetic transfer exhibits a structure that does not fit into standard cohomological frameworks. The five key features are:

1. The cocycle is *twisted*: the internal error is amplified exponentially by the external context.
2. The cochain complex is *curved*: $d^2 \circ d^1 = [\delta, \cdot] \neq 0$.
3. The curvature is *non-trivial*: $[\delta] \neq 0$ in $H^2 \cong \mathbb{R}$, proved by surdetermination.
4. The parameter space is *fibered*: each base α defines its own cohomology, with incompatible fibers.
5. The construction is *arithmetic*: the curvature arises from the elementary map $n \mapsto \alpha^n$, not from geometry or algebra.

The formula $d^2 \circ d^1 = [\delta, \cdot]$ places the construction in the theory of curved complexes [7], but with a new feature: the curvature element has a number-theoretic origin and carries computational information (the growth of δ controls the invertibility of the transfer; see [13]).

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